AI Governance Gaps in Wearable Health Data:

A Parkinson's Tremor Detection Case Study

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Artificial intelligence is becoming increasingly common in healthcare, especially through the use of wearable devices such as smartwatches and fitness trackers. The FDA now lists more than 1,600 AI-enabled medical devices authorized for marketing in the United States (U.S. Food and Drug Administration, n.d.). Wearables continuously collect physiological and movement-based data, including heart rate, sleep activity, and accelerometer readings. Researchers and healthcare organizations are now using this data to train machine learning models capable of detecting diseases, monitoring symptoms, and supporting clinical decision-making. One area where this has become especially important is Parkinson's disease monitoring. Parkinson's disease is a neurological disorder that affects movement and often produces tremors, rigidity, and motor impairment. Because wearable sensors can record movement outside the clinic, they create an opportunity for continuous monitoring of symptoms at home (Sica et al., 2021).

Although wearable healthcare AI systems offer significant benefits, they also introduce governance challenges that are often overlooked. Current discussions around AI governance in healthcare typically focus on issues such as fairness, privacy, transparency, and accountability. While these concerns are important, they do not fully address technical factors that directly affect how reliable these systems are in practice. Small differences in how data is collected, processed, and structured can significantly impact how machine learning models behave. As a result, a model that appears accurate in one environment may fail when deployed in another. Hatherley

(2025) describes this as dataset shift, a mismatch between the data a model was developed on and the data it is later used on.

This paper argues that current AI governance frameworks fail to adequately address how differences in data preprocessing and dataset composition impact model reliability. Using a Parkinson's tremor detection research project as a case study, this paper demonstrates how technical inconsistencies in wearable sensor data can create reliability risks that the frameworks reviewed here do not fully capture.

Artificial intelligence systems in healthcare rely heavily on large datasets to identify patterns and make predictions. In wearable healthcare systems, these datasets often consist of time-series sensor data collected from accelerometers and motion-tracking devices. Researchers use this information to detect movement abnormalities associated with diseases such as Parkinson's disease. However, wearable health data differs from traditional clinical datasets because it is highly variable and often collected in uncontrolled real-world environments. Factors such as device type, sensor quality, user behavior, and sampling frequency can all influence the structure of the data.

Research on bias in medical AI has shown that problems can emerge and compound at multiple stages of the machine learning pipeline, including data collection, labeling, model development and evaluation, and deployment (Cross et al., 2024). The same research notes that a model's performance can deteriorate when it is applied to data outside the cohort it was trained on. This means that even if two datasets are designed to measure similar physical activities, differences in how the data is structured may produce very different outcomes when processed by the same model. Fairness is often discussed in terms of demographic bias, but technical inconsistencies in datasets themselves may also create unequal or inaccurate outcomes. This

issue became especially apparent during my research on Parkinson's tremor detection using wearable accelerometer data. The project focused on building a Long Short-Term Memory (LSTM) model capable of detecting tremor-related movement patterns from time-series data. Two publicly available datasets were analyzed: PAMAP2 (Reiss, 2012) and Capture-24 (Chan et al., 2024). PAMAP2 contains recordings of 9 subjects performing 18 physical activities while wearing inertial sensors on the wrist, chest, and ankle, along with a heart rate monitor, and the subjects followed a set protocol. Capture-24 contains accelerometer data from 151 adults who wore a wrist device for about 24 hours in their everyday lives. The two datasets were collected in very different ways, and they differed significantly in structure and preprocessing methods.

Several technical inconsistencies emerged during the project. One major issue involved feature mismatches between datasets. The training data contained nine features, while the validation data contained only seven. This caused evaluation failures because the model expected a different input structure than what was provided during testing. There were also scaling differences between datasets. One dataset ranged approximately between positive and negative one, while another ranged closer to positive and negative five. Although these differences may appear minor, they significantly affected how the model interpreted movement patterns. The project also relied on time-series segmentation using 256-sample windows with overlapping windows during preprocessing. Missing values introduced additional complications because Fast Fourier Transform (FFT) analysis could not be performed when null values existed in the dataset. As a result, interpolation had to be introduced during preprocessing to fill missing data points. These preprocessing decisions directly influenced how the model interpreted the underlying motion signals.

The key finding from the project was that the same machine learning model produced different results depending on how the data was prepared. Preprocessing steps such as normalization, scaling, interpolation, and feature selection directly changed model behavior. In some situations, mismatched features caused the model to fail entirely. In others, different preprocessing pipelines led to noticeable differences in accuracy. One experiment achieved approximately 0.82 accuracy when trained on one dataset and tested on another under specific preprocessing conditions. However, this performance depended heavily on the exact preprocessing pipeline used during evaluation. When preprocessing conditions changed, the model's behavior and performance also changed significantly. In this project, reported accuracy depended more on preprocessing conditions than on the robustness of the model itself, and I think that is a limitation worth looking for in other healthcare AI systems.

These findings highlight a broader governance issue within healthcare AI. The World Health Organization (WHO) and the Food and Drug Administration (FDA) have both published guidance for AI in healthcare, and each covers more ground than fairness and privacy alone. WHO's report calls for health data to be of high quality, representative, and harmonized, and says that data from wearable technologies should be regulated (WHO, 2021, pp. 142–143). It also notes that a recent comparison of deep-learning algorithms and health-care professionals in medical imaging found that few studies present externally validated results, and it asks whether AI trained for use in one context can be used accurately and safely in a different region or context (WHO, 2021, p. 6). The FDA describes a risk-based approach that follows a device across its total product life cycle (U.S. Food and Drug Administration, n.d.), and the Good Machine Learning Practice principles it points to, published by the International Medical Device Regulators Forum (IMDRF), go further. They call for datasets that represent the intended patient

population (principle 3), training and test sets that are independent of one another with external validation proportionate to risk (principle 4), testing that is independent of the training data (principle 8), and monitoring for dataset drift after deployment (principle 10) (IMDRF, 2025). These are important commitments to oversight and patient safety.

However, in the WHO report, the FDA overview page, and the IMDRF principles I reviewed for this paper, these commitments stay at the level of principles. They ask for independent test data and for external validation in proportion to risk, but I did not find detailed standards for preprocessing consistency, or a requirement to show that a model holds up across independent datasets that were collected and processed in different ways. This creates a gap between technical reality and governance policy. A model may satisfy regulatory expectations under controlled testing conditions while still failing when exposed to new datasets collected under different circumstances, which is the pattern my project showed on a small scale. If governance does not ask developers to show that a model stays reliable across independent datasets with different preprocessing pipelines, a healthcare AI system may look trustworthy even though it is sensitive to technical inconsistencies.

This problem becomes especially serious in wearable healthcare systems because the data is continuously collected and interpreted in real time. If a wearable AI system produces inconsistent predictions due to preprocessing differences, users and clinicians may not realize that the outputs are unreliable. In the context of Parkinson's disease, this could lead to inaccurate symptom monitoring, delayed treatment adjustments, or incorrect clinical interpretations. Sample sizes in this area are also small. Several of the tremor-focused studies described in a systematic review of home monitoring included only one to ten patients (Sica et al., 2021), so there is little evidence yet about how well these models hold up across different people and settings.

Research on bias in medical AI supports the idea that technical inconsistencies can create unequal outcomes. Cross et al. (2024) describe how a model that performs well on the data it was built on can perform worse, and differently across patient groups, once it is deployed on a population that differs from its training data. Bias in machine learning systems is therefore not limited to demographic representation. My project suggests that differences in feature engineering, preprocessing methods, and dataset composition can also affect model reliability, which means two patients performing similar physical movements could receive different AI-generated assessments depending on how their data was collected or processed. In wearable healthcare systems, these risks are amplified because devices are often used outside controlled laboratory environments, where sensor placement, motion patterns, environmental noise, and user behavior all vary.

Another governance concern involves transparency. WHO suggests that transparency about the provenance of the data behind an AI system, and about how that data was processed, would let others validate the technology independently and increase the trust of its users (WHO, 2021, p. 108). The IMDRF principles similarly ask that users be given the characteristics of the data used to train and test a model and the study methodology (IMDRF, 2025, principle 9), but they do not name preprocessing steps such as normalization or interpolation. My project shows why this detail matters. Without knowing the normalization methods, feature selection, interpolation techniques, or dataset composition behind a reported accuracy, it is hard for researchers and clinicians to judge whether the result is reproducible or generalizable. This is why governance discussions need to move from abstract ethical principles toward technical evaluation standards. Ethical AI is not only about fairness and privacy. It is also about making sure systems behave reliably across different conditions and populations.

Several potential solutions could help address these gaps. One important improvement would involve requiring cross-dataset validation during model evaluation. Existing principles already call for external validation in proportion to risk, so this would make that expectation concrete for wearable data. Rather than testing a model on only one controlled dataset, developers could be required to evaluate models across multiple independent datasets collected under different conditions. This would provide a stronger measure of generalizability and robustness. Another possible solution involves standardizing preprocessing pipelines for wearable healthcare data. Standardization could reduce inconsistencies introduced through scaling differences, feature mismatches, or interpolation techniques. If organizations established clearer preprocessing standards, models would be less likely to behave unpredictably across datasets.

Transparency requirements should also be strengthened. Researchers and companies should clearly document how data is processed before training and evaluation. This includes reporting normalization methods, feature engineering steps, missing-data handling techniques, and validation conditions. Greater transparency would improve reproducibility and allow regulators and clinicians to better evaluate the reliability of healthcare AI systems.

Finally, stronger collaboration between policymakers, healthcare professionals, and data scientists is necessary. Governance frameworks are often written at a high conceptual level without fully accounting for technical realities encountered during machine learning development. Researchers working directly with wearable datasets can provide important insights into how preprocessing decisions affect model behavior. Integrating technical expertise into governance discussions would help create standards that better reflect real-world AI deployment challenges.

The Parkinson's tremor detection case study demonstrates that even small preprocessing differences can lead to major inconsistencies in machine learning performance. Feature mismatches, scaling differences, and interpolation decisions all influenced model reliability during experimentation. These technical issues highlight an important limitation in the governance frameworks I reviewed: they do not fully address how data preprocessing and dataset variability affect real-world system performance. As healthcare increasingly adopts wearable AI systems, governance frameworks must evolve beyond broad ethical principles and incorporate stronger technical standards. Without improved validation requirements, preprocessing transparency, and cross-dataset testing, healthcare AI systems may appear accurate in controlled settings while failing in real-world environments. Reliable AI governance must therefore consider not only the model itself, but the entire data pipeline that shapes how the model interprets human health data.

References

Chan, S., Yuan, H., Tong, C., Acquah, A., Schonfeldt, A., Gershuny, J., & Doherty, A. (2024). CAPTURE-24: A large dataset of wrist-worn activity tracker data collected in the wild for human activity recognition. *Scientific Data*, 11, 1135. https://doi.org/10.1038/s41597-024-03960-3

Cross, J. L., Choma, M. A., & Onofrey, J. A. (2024). Bias in medical AI: Implications for clinical decision-making. *PLOS Digital Health*, 3(11), e0000651. https://doi.org/10.1371/journal.pdig.0000651

Hatherley, J. (2025). A moving target in AI-assisted decision-making: Dataset shift, model updating, and the problem of update opacity. *Ethics and Information Technology*, 27, Article 20. https://doi.org/10.1007/s10676-025-09829-2

Reiss, A. (2012). *PAMAP2 physical activity monitoring* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5NW2H

Sica, M., Tedesco, S., Crowe, C., Kenny, L., Moore, K., Timmons, S., Barton, J., O'Flynn, B., & Komaris, D.-S. (2021). Continuous home monitoring of Parkinson's disease using inertial sensors: A systematic review. *PLOS ONE*, 16(2), e0246528. https://doi.org/10.1371/journal.pone.0246528

U.S. Food and Drug Administration. (n.d.). *Artificial intelligence-enabled medical devices*. Retrieved October 16, 2026, from https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-software-medical-device

World Health Organization. (2021). *Ethics and governance of artificial intelligence for health: WHO guidance*. https://www.who.int/publications/i/item/9789240029200